```
# Part 1: Arithmetic Mean Method
machines = {
    "A": [3, 2, 4],
    "B": [6, 7],
    "C": [5, 4, 6]
}

Q_values = {}
for machine, rewards in machines.items():
    Q_values[machine] = sum(rewards) / len(rewards)
    print(f"Machine {machine}")
    print(f"Rewards: {rewards}")
    print(f"Q({machine}) = {Q_values[machine]:.2f}")

greedy_choice = max(Q_values, key=Q_values.get)
print("Greedy Selected Machine:", greedy_choice)


# Part 2: Incremental Update Method
Q_inc = {m: 0 for m in machines}
N = {m: 0 for m in machines}

for machine, rewards in machines.items():
    for r in rewards:
        N[machine] += 1
        Q_inc[machine] = Q_inc[machine] + (r - Q_inc[machine]) / N[machine]

print("\nIncremental Update Results:")
for machine in machines:
    print(f"Q({machine}) = {Q_inc[machine]:.2f}")
```

```
Machine A
Rewards: [3, 2, 4]
Q(A) = 3.00
Machine B
Rewards: [6, 7]
Q(B) = 6.50
Machine C
Rewards: [5, 4, 6]
Q(C) = 5.00
Greedy Selected Machine: B

Incremental Update Results:
Q(A) = 3.00
Q(B) = 6.50
Q(C) = 5.00
```

```
# Incremental Update Method for Multi-Armed Bandit

machines = {
    "A": [3, 2, 4],
    "B": [6, 7],
    "C": [5, 4, 6]
}

# Initialize Q and N
Q = {m: 0 for m in machines}
N = {m: 0 for m in machines}

# Process rewards incrementally
for machine, rewards in machines.items():
    for r in rewards:
        N[machine] += 1
        Q[machine] = Q[machine] + (r - Q[machine]) / N[machine]
        print(f"After playing {machine}, reward={r}")
        print(f"N({machine})={N[machine]}, Q({machine})={Q[machine]:.2f}")
        # Identify greedy choice at this step
        greedy_choice = max(Q, key=Q.get)
        print("Greedy Selected Machine:", greedy_choice)
        print("-" * 40)

# Final results
print("\nFinal Q values:")
for machine in machines:
    print(f"Q({machine}) = {Q[machine]:.2f}")

greedy_choice = max(Q, key=Q.get)
print("Final Greedy Selected Machine:", greedy_choice)
```

```
After playing A, reward=3
N(A)=1, Q(A)=3.00
Greedy Selected Machine: A
----------------------------------------
After playing A, reward=2
N(A)=2, Q(A)=2.50
Greedy Selected Machine: A
----------------------------------------
After playing A, reward=4
N(A)=3, Q(A)=3.00
Greedy Selected Machine: A
----------------------------------------
After playing B, reward=6
N(B)=1, Q(B)=6.00
Greedy Selected Machine: B
----------------------------------------
After playing B, reward=7
N(B)=2, Q(B)=6.50
Greedy Selected Machine: B
----------------------------------------
After playing C, reward=5
N(C)=1, Q(C)=5.00
Greedy Selected Machine: B
----------------------------------------
After playing C, reward=4
N(C)=2, Q(C)=4.50
Greedy Selected Machine: B
----------------------------------------
After playing C, reward=6
N(C)=3, Q(C)=5.00
Greedy Selected Machine: B
----------------------------------------

Final Q values:
Q(A) = 3.00
Q(B) = 6.50
Q(C) = 5.00
Final Greedy Selected Machine: B
```